

Generative AI - Course Outline

Duration

20 Hours (3 Hours/Day)

Course Overview

This hands-on program provides participants with practical knowledge of **Generative AI, Large Language Models (LLMs), prompt engineering, AI APIs, conversational AI, semantic search, and multimodal AI**. Participants will learn how to leverage modern AI technologies to build intelligent applications, automate workflows, and enhance productivity.

The program helps participants:

- Understand **Generative AI, NLP, Transformers, and LLM fundamentals**
- Apply effective **prompt engineering techniques**
- Integrate AI capabilities into applications using APIs
- Build **AI-powered chatbots and assistants**
- Implement **semantic search using embeddings** and vector databases
- Explore **multimodal AI** and image generation capabilities
- Apply **Responsible AI** principles and best practices

By the end of the program, participants will be able to develop practical AI-powered solutions for real-world business and enterprise use cases.

Pre-requisites

- Basic understanding of AI/ML concepts
- Familiarity with Python programming
- Basic understanding of APIs and data handling
- Awareness of cloud platforms (optional)
- Interest in AI applications and automation

Labs

- Python (3.12+)
- VS Code / Jupyter Notebook
- OpenAI or similar API access
- Internet connectivity

Course Outline (20 Hours)

Module 1: Generative AI Foundations & Ecosystem

Topics

- AI vs ML vs Deep Learning vs Generative AI
- Evolution of Artificial Intelligence
- What is Generative AI?
- Types of Generative AI
 - Text Generation
 - Image Generation
 - Audio Generation
 - Video Generation
 - Code Generation
- Industry Use Cases and Business Applications
- Foundation Models and AI Ecosystem
- Overview of Popular Models - GPT, Claude, Gemini, Llama, Mistral, etc.
- Introduction to Responsible Use of AI

Hands-on

- Exploring Generative AI tools
- Content generation using LLMs
- Comparing outputs from different AI models

Module 2: NLP, Transformer Architecture & LLM Fundamentals

Topics

- Introduction to Natural Language Processing (NLP)
- NLP Tasks and Applications
- Evolution of NLP Systems
- Tokens and Language Representation
- Embeddings Fundamentals
- Challenges of Traditional NLP Models
- Introduction to Transformer Architecture
- Attention Mechanism Overview
- Encoder and Decoder Concepts
- Understanding Large Language Models (LLMs)
- Context Windows and Token Management
- Prompt-Response Lifecycle

Hands-on

- Text processing exercises
- Exploring tokenization
- Working with LLM interfaces

Module 3: Prompt Engineering

Topics

- Principles of Effective Prompt Design
- Prompt Components and Structure
- Zero-shot Prompting
- Few-shot Prompting
- Role-based Prompting
- Instruction-based Prompting
- Structured Output Generation
- Prompt Optimization Techniques
- Prompt Evaluation and Refinement

Hands-on

- Prompt engineering exercises
- Prompt optimization techniques
- Generating structured outputs

Module 4: OpenAI APIs & AI Application Integration

Topics

- API Fundamentals
- OpenAI and LLM API Ecosystem
- Authentication and API Access
- Model Selection and Parameters
- Integrating AI into Applications
- Request and Response Handling
- Error Handling and Best Practices
- AI Application Design Considerations

Hands-on

- Calling LLM APIs using Python
- Building an API-driven content generation application

Module 5: Conversational AI & Chatbot Development (3 Hours)

Topics

- Introduction to Conversational AI
- Chatbot Design Principles
- Conversation Flow Management
- Context and Memory Handling
- Prompt Orchestration for Chatbots

- Building AI Assistants
- Enterprise Chatbot Use Cases
- Chatbot Deployment Considerations

Hands-on

- Building an AI-powered chatbot
- Enhancing chatbot responses using prompts

Session 6: Retrieval & Multimodal AI

Topics

- Embeddings and Vector Representations
- Similarity Search Concepts
- Vector Databases Overview
- Semantic Search Architecture
- Information Retrieval with Embeddings
- Introduction to Multimodal AI
- Text-to-Image Generation Concepts
- Diffusion Models Overview
- Image Generation Tools and Platforms
- Prompting Techniques for Image Generation

Hands-on

- Semantic Search Demonstration
- Working with Embeddings
- Image Generation Exercises
- Prompt Optimization for Images

Session 7: Responsible AI & Mini Project

Topics

- Bias and Fairness in AI Systems
- Hallucinations and Reliability Challenges
- Privacy, Security and Compliance Considerations
- AI Governance Principles
- Best Practices for Enterprise AI Adoption
- Future Trends in Generative AI

Mini Project

- Build a simple AI-powered application
 - Chatbot OR
 - Content Generator
- Project Demonstration
- Review and Feedback